

Python + SQL Case Study: Urban Ghost Kitchen Analytics

Business Context

A food-delivery company operates several ghost kitchens across a city. Unlike traditional restaurants, these kitchens have no dine-in area and depend entirely on delivery orders.

Management noticed that some kitchens receive many orders but still generate poor profits because of long delivery distances, discounts, cancellations, and low-value orders.

Your job is to identify where the operational and financial problems are occurring.

Dataset

You will work with 4 CSV files:

1. `customers.csv`

column	description
customer_id	Unique customer ID
city	Customer city
signup_date	Registration date
customer_segment	New / Regular / Premium

2. orders.csv

column	description
order_id	Unique order ID
customer_id	Customer ID
kitchen_id	Ghost kitchen ID
order_date	Order date
order_value	Gross order value
discount	Discount given
delivery_fee	Delivery fee
delivery_distance_km	Delivery distance
delivery_time_min	Delivery time
order_status	Delivered / Cancelled / Refunded

3. kitchens.csv

column	description
kitchen_id	Unique kitchen ID
kitchen_name	Kitchen name
zone	Operating zone
monthly_rent	Monthly kitchen rent
staff_count	Number of employees

4. menu_items.csv

column	description
item_id	Unique item ID
order_id	Order ID
item_name	Food item
category	Pizza / Burger / Rice / Dessert etc.
quantity	Quantity ordered
unit_price	Price per item

Your Business Questions

Solve these using SQL + Python.

SQL Tasks

1. Find the top 5 kitchens by total revenue.
2. Calculate each kitchen's:
 - total orders
 - delivered orders
 - cancelled orders
 - cancellation rate
3. Find the average delivery time for each kitchen.
4. Find the average order value by customer segment.
5. Identify kitchens where:
 - cancellation rate > 8%
 - average delivery time > 35 minutes
6. Find the top 10 customers by total spending.
7. Find the most popular food category in each zone.
8. Calculate monthly:
 - gross revenue
 - discounts
 - delivery fees
 - net revenue

Python Tasks

Load the SQL output/results into Pandas and perform analysis.

1. Data Quality

Check:

- missing values
- duplicate IDs
- incorrect datatypes
- negative order values
- negative discounts
- impossible delivery distances
- impossible delivery times

2. Feature Engineering

Create:

- net_order_value
- discount_rate

- `delivery_efficiency`
- `customer_lifetime_value`

Suggested formulas:

- `net_order_value = order_value - discount + delivery_fee`
- `discount_rate = discount / order_value`
- `delivery_efficiency = order_value / delivery_time_min`
- For `customer_lifetime_value`, use the customer's cumulative net spending.

Final Analysis

Answer these 5 questions:

Q1 Which kitchen generates the most revenue?

Q2 Which kitchen has the biggest operational problem?

Don't simply use revenue. Consider:

- cancellation rate
- delivery time
- order volume

Q3 Which customer segment is most valuable?

Compare:

- number of customers
- total revenue
- average order value
- average orders per customer

Q4 Which food category should management investigate?

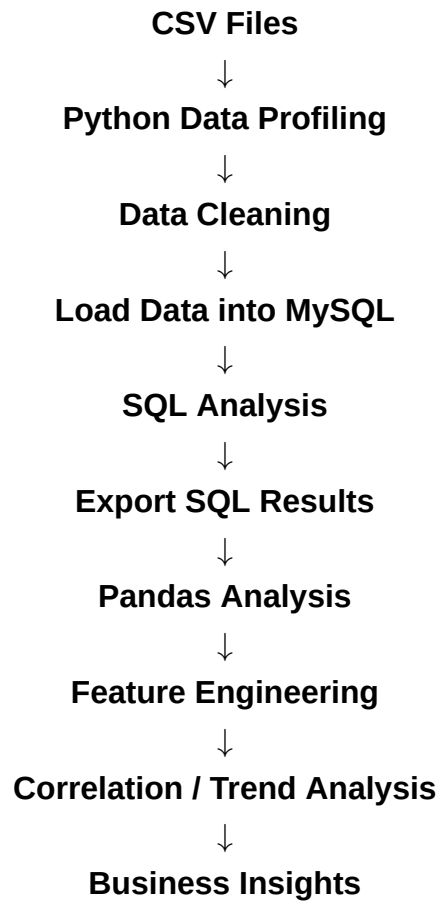
Consider:

- order quantity
- revenue
- average selling price

Q5 Does a higher discount appear to be associated with higher order values?

Use Python to investigate the relationship.

Expected Workflow



Constraint

Do not use Power BI for this case study.

Your Final Output Should Contain:

1. Data Profiling
2. Data Cleaning
3. SQL Queries
4. Python Analysis
5. 5 Business Findings
6. 3 Business Recommendations

Difficulty Metrics

- Difficulty: Beginner → Intermediate
- Estimated time: 2–3 hours
- Main skills tested: Pandas, SQL JOIN, GROUP BY, CASE, CTE, window functions, aggregation, feature engineering, correlation, and business reasoning.